$$= 16ab \int_0^1 \left[\int_0^1 x^2 \, dx \right] y^2 \, dy$$

$$= 16ab \int_0^1 \frac{1}{3} y^2 \, dy$$

$$= 16ab \times \frac{1}{3} \times \frac{1}{3} = \frac{16ab}{9} \quad \rightarrow 1$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) \, dx$$

$$= \int_0^1 x \times 4ax \, dx = 4a \int_0^1 x^2 \, dx$$

$$= 4a\left(\frac{1}{3}\right) = \frac{4a}{3}$$

$$E(Y) = \int_{-\infty}^{\infty} y f(y) \, dy$$

$$= \int_0^1 y \times 4by \, dy$$

$$= 4b \int_0^1 y^2 \, dy$$

$$= 4b\left(\frac{1}{3}\right) = \frac{4b}{3}$$

$$E(X)\,E(Y) = \frac{4a}{3} \times \frac{4b}{3}$$

$$= \frac{16ab}{9} \quad \rightarrow 2$$

From 1 and 2,

$$E(XY) = E(X)\,E(Y)$$

9.8 Moments

Another approach helpful to find the summary measures for probability distribution is based on the 'moments'.We will discuss two types of moments.

(i) Moments about the origin. (Origin may be zero or any other constant say A). It is also called as raw moments.

(ii) Moments about the mean is called as central moments.